

Olist Brazilian E-Commerce — SQL Data Analysis

Data Overview

The Olist Brazilian E-Commerce dataset contains 9 tables covering customers, orders, products, payments, reviews, sellers, and other transaction-related information. The orders table contains 99,441 orders from September 4, 2016 to October 17, 2018. Most orders were successfully delivered, with 96,478 delivered orders, while a smaller number were shipped, canceled, unavailable, or in other statuses. The early months of the dataset contain very low transaction activity, so percentage changes during these periods should be interpreted carefully.

Business Insights

1. Revenue Growth

Revenue showed strong growth over the dataset period. Monthly revenue increased from 138,488.04 in January 2017 to 1,115,004.18 in January 2018, showing substantial year-over-year growth. However, the earliest months had very low transaction activity, which produced unusually large month-over-month percentage changes. This means the later, more established months provide a more reliable view of the business trend.

2. Customer Retention

Most customers purchased only once. The customer behavior analysis found 92,851 one-time buyers compared with only 2,974 repeat buyers, meaning approximately 96.9% of customers were one-time buyers and only 3.1% were repeat buyers. This suggests that although Olist was able to attract many customers, converting existing customers into repeat buyers represents a significant growth opportunity.

3. Product Category Performance

Beauty and Health was the highest-revenue product category. It generated 39,010.07 in product revenue and accounted for 10.79% of total category revenue. However, because the leading category represents only about one-tenth of total revenue, sales are distributed across multiple product categories rather than being heavily dependent on a single category.

Recommendations

1. Improve Customer Retention

Olist should focus on increasing repeat purchases through targeted post-purchase campaigns, personalized product recommendations, and incentives for returning customers. Since only about 3.1% of customers made repeat purchases, even a modest improvement in customer retention could create additional revenue from the existing customer base.

2. Strengthen High-Performing Categories While Maintaining Diversity

Olist should continue supporting strong categories such as Beauty and Health while identifying opportunities to grow other high-performing categories. Because the top category contributes only 10.79% of category revenue, the company can maintain a diversified product strategy rather than relying too heavily on one category.